

Unified Case on Gamification of Practical Assignments

Al Sana 5-Hour Practical Hackathon

Business Task Quality Rating and Open Team Selection

The team must build a working MVP that helps a business representative turn an initial description of their need or task into a complete card, evaluates how well it is filled in, and publishes the task in an open catalog. Students choose interesting tasks themselves and submit proposals, and the business representative decides whom to continue working with.

Parameter	Condition
Duration	5 hours, including demo preparation
Team	3–5 participants
Format	A web application or interactive prototype with working logic
Main result	An end-to-end scenario from a business task draft to publication, a proposal and team selection

Key principle

The main gamification targets the business: the more complete and useful the task description, the higher its rating and catalog position. The system may recommend tasks to students but does not assign teams.

1 Task for Participants

Develop an MVP of a platform where a business representative describes their need, problem or task. The system helps fill in missing details, calculates a readiness rating and publishes the card in a shared catalog. Any student team can view the task and submit a proposal. The business compares the proposals itself and selects one, several or none of the teams.

2 End-to-End User Scenario

Step	Action	Result
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1	Draft	The business representative enters a short description of the need, problem or task.
2	Clarification	The system identifies missing details and asks at least three relevant questions.
3	Card	The answers are turned into an editable card, which the business representative confirms.
4	Rating	The system awards points for completeness and shows which details would raise the task's rating.
5	Catalog	The confirmed task is published in the shared pool at a position matching its rating.
6	Student choice	Any team browses the catalog or recommendations and submits its proposal.
7	Business choice	The business compares proposals and itself selects one, several or none of the teams.
8	Result	After a confirmed milestone, the selected team receives points for actual progress.

3 Required Functionality

Module	Minimum requirement
Task builder	Free-form input; completeness check; at least 3 clarifying questions; editable card; manual confirmation.
Task card	Title, context, need, users, data, constraints, expected result, success criteria, contact and interaction format.
Task rating	Score from 0 to 100; breakdown of awarded points; list of missing details; recalculation after editing.
Shared catalog	All published tasks are available to all teams; sorting by rating; filters by topic and readiness level.
Team proposals	Any team can submit a solution idea, a plan and a link to a prototype. The number of proposals is not limited.
Business choice	A list of proposals and manual actions to select or reject. Automatic team assignment is prohibited.

4 Core Gamification Mechanics

The rating reflects not the company's fame but the readiness of a specific business task for work with students. Points are awarded only for filled-in and confirmed fields. Recommended scale:

Indicator	Weight	How to score
Context and need	20 points	It is clear what is happening now and what needs to change
Data and materials	20 points	Available data, examples or sources are specified
Expected result	15 points	A specific result of the team's work is described
Success criteria	15 points	There are measurable signs that the solution is accepted
Constraints	10 points	Deadlines, technologies, access or other boundaries are specified
Users	10 points	It is clear who the solution is being built for
Connection with the business	10 points	There is a contact, a consultation format and a feedback procedure

Task Readiness Levels

- 0–39 points — draft: the task is visible in the catalog but marked as needing clarification;
- 40–69 points — workable: students can submit proposals, and the system may recommend the task;
- 70–89 points — ready: the task gets a higher position in the catalog;
- 90–100 points — priority: the task is fully ready for work and is highlighted in the catalog.

A low rating does not hide the task or prevent proposals. It shows students how realistic it is to start work without further clarification. The rating is recalculated after each confirmed addition.

5 Use of Artificial Intelligence

The MVP must include at least one substantive AI function. It is recommended to use AI to analyze the completeness of the description and generate clarifying questions, or to convert answers into a task card. If an external API is unavailable, a local stub is acceptable, but the team must show the prompt, the input and output format, and the handling of an invalid response.

- AI must not add facts that the user did not provide.
- The generated text is edited and confirmed by a human before publication.

- AI may recommend tasks to students based on interests and skills but does not restrict the shared catalog.
- AI does not choose a team on behalf of the business and does not assign contractors automatically.
- Personal and sensitive attributes of participants are not used.

6 Source Data

The organizers may provide a ready-made JSON or CSV. If no dataset is provided, the team creates its own synthetic data in the specified minimum volume.

Dataset	Minimum volume	Required fields
Task drafts	5 descriptions	Text, industry and details of varying completeness
Task cards	5 cards	All rating fields and the current readiness score
Team profiles	5 profiles	Name, interests, skills and technologies
Proposals	5 proposals	Team, solution idea, plan, timeline and link

7 Scope Limits

Within five hours, the following are not required:

- full registration, password recovery and a complex role model;
- real-time chat, notifications, calendar and file storage;
- training your own ML model or a vector database;
- mobile adaptation and production infrastructure;
- a full project tracker — a proposal and the business's decision on it are enough for the demo.

8. 5-Hour Work Plan

Time	Stage result
0–30 minutes	Roles assigned, stack chosen, data schema and end-to-end scenario agreed

30–100 minutes	Input, clarifying questions and the editable business task card work
100–165 minutes	Rating formula, readiness levels and improvement recommendations work
165–225 minutes	Shared catalog, team proposal and manual business choice work
225–270 minutes	End-to-end scenario integrated, validation and error handling added
270–300 minutes	Test data, README and a demo of up to 5 minutes prepared

9 Evaluation Criteria

Criterion	Points	What is checked
End-to-end scenario	20	A draft goes through publication, a proposal and the business decision.
Card quality	15	Questions are relevant; the card is complete and editable.
Business gamification	25	The rating is transparent, recalculated and affects position.
Catalog and proposals	15	All tasks are available; the business selects the team manually.
AI function	10	The result is useful, structured and contains no invented data.
Technical quality	10	There is a launch procedure, validation, structure and instructions.
Demo	5	A complete working scenario is shown in 5 minutes.

Maximum 100 points

10 Submission Deliverables

- A link to the repository or an archive with the source code.
- A working MVP that can be launched following the instructions.
- A README with the architecture, task rating formula, catalog rules and test scenarios.
- A demo of up to 5 minutes on one prepared example.

11 Mandatory Demo

At the defense, participants enter a weak business task description, improve it and show the rating increase. The task is then published in the shared catalog, a student team independently submits a proposal, and the business manually accepts or rejects it. All transitions must work in the MVP, not on slides.